Universidade do Minho
Escola de Engenharia
Departamento de Informática

KNIME

Pre-Processing + Linear Regression

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA

Inteligência Artificial para as Telecomunicações

2025/2026

Objectives

- The primary goal of this task is **to create a comprehensive and efficient workflow** tailored specifically for creating a Linear Regression model.

- The goal is to create a linear regression model to predict the price of lego sets.

Pre-processing

- Data Cleaning: Identifying and handling missing values, correcting inconsistencies, and removing duplicates to ensure data quality.
- Data Transformation: Applying necessary transformations, such as scaling, normalization, and encoding categorical variables, to standardize the dataset.
- Data Integration: Combining or merging multiple datasets to create a unified dataset that is ready for analysis.
- Feature Engineering and Selection: Creating new features, selecting important ones, or reducing dimensionality to improve model performance and computational efficiency.

Dataset

Lego_sets

- A .csv file

LEGO is a brand of building blocks and they are usually sold in sets to build a specific object.

This dataset contains data about these sets, including the number of pieces in each set, selling price, suggested age, among others.

Lego Sets

- **age:** The suggested age the set should be used by. It can assume values like the range of 6-12 years, 10+ years or 12+ years;
- **list price**: The price of the set in dollars. Some values can be 9.99 dollars, 19.99 dollars, 29.99 dollars, etc;
- **num reviews**: Number of reviews per set;
- **piece count**: Number of pieces in that LEGO set;
- **play star ratings**: The ratings of the set;
- **review difficulty**: Level of difficulty of the set. This attribute can assume the values ”Very Easy”, ”Easy”, ”Average”, ”Challenging” and ”Very Challenging”;
- **star ratings**: The ratings of the set;
- **theme name**: Which theme the set belongs to. The themes can be Architecture, BOOST, BrickHeadz, etc;
- **val star rating**: The ratings of the set;
- **country**: The name of the country the set comes from. The countries can be US, Australia, etc.

KNIME Analytics Platform

Goal:

- Understand **Linear Regression**
- Develop a **Linear Regression model** capable of **predicting the price** of a LEGO set, given a set of features available in a dataset
- Analyze the measures used to **evaluate the model**

Linear Regression

Linear Regression is used when we want to **predict** the **value of one independent variable** based on the **value of another dependent variable**.

This helps determine whether an independent variable **does a good job of predicting** the dependent variable or **which independent variable plays a significant role** in predicting the dependent variable.

Linear Regression

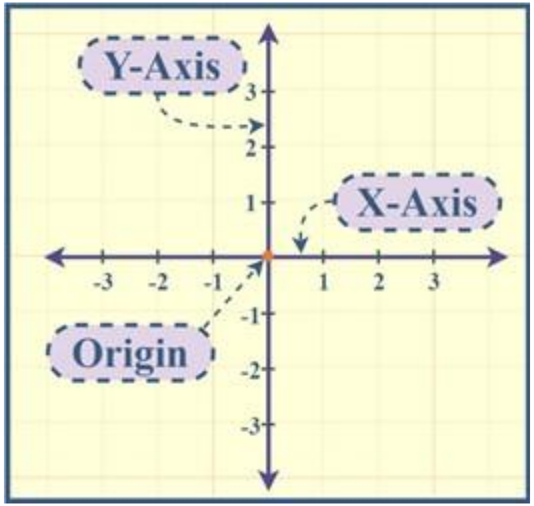
- **Independent variable**: a variable (often denoted by ***x***) whose **variation does not depend** on that of another, **prediction variable** used to be estimated;
- **Dependent variable**: a variable (often denoted by ***y***) whose **value depends** on that of another, **target variable** to be predicted.

Linear Regression

- **Slope: line angle**, identified as ***m*** or ***β1***;
- **Intercept**: where the **function crosses the y-axis**, identified as **b** or ***β0***.

$$y = \beta 0 + \beta 1 X + \epsilon$$

Linear Regression



$$y = \beta 0 + \beta 1 X + \epsilon$$

Linear regression **finds the best fitting line** through your data by looking for the regression coefficient (**β1)** that **minimises the total error** (**ε**) of the model.

Exercise:

- **Problem:** Development of a Machine Learning Model capable of **predicting the price of a particular LEGO set**.
- **Upload the pre-processing workflow developed for the LEGO dataset.**

Dataset: Table with information on LEGO sets, containing:

- '**age**': What age category do you belong to
- '**list_price**': set price (in $)
- '**num_reviews**': Number of revisions per set
- '**piece_count**': Number of pieces in that LEGO set
- '**play_star_ratings**': ratings
- '**review_difficulty**': level of difficulty of the set
- '**star_rating**': ratings
- '**theme_name**': Which theme it belongs to
- '**val_star_rating**': ratings
- '**country**': Country name

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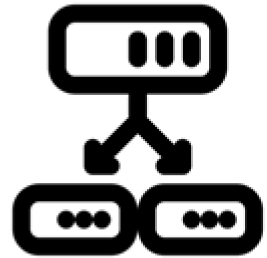
Exercise:



First Step: Partition the data

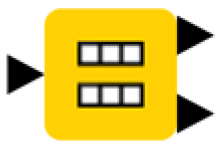
- Partitioning the data into **training** and **testing sets** is a fundamental step in model development;
- **Train a model on one subset** of the data (training set);
- **Evaluate** how well the trained model **generalizes to unseen data** (test set).

ISLab

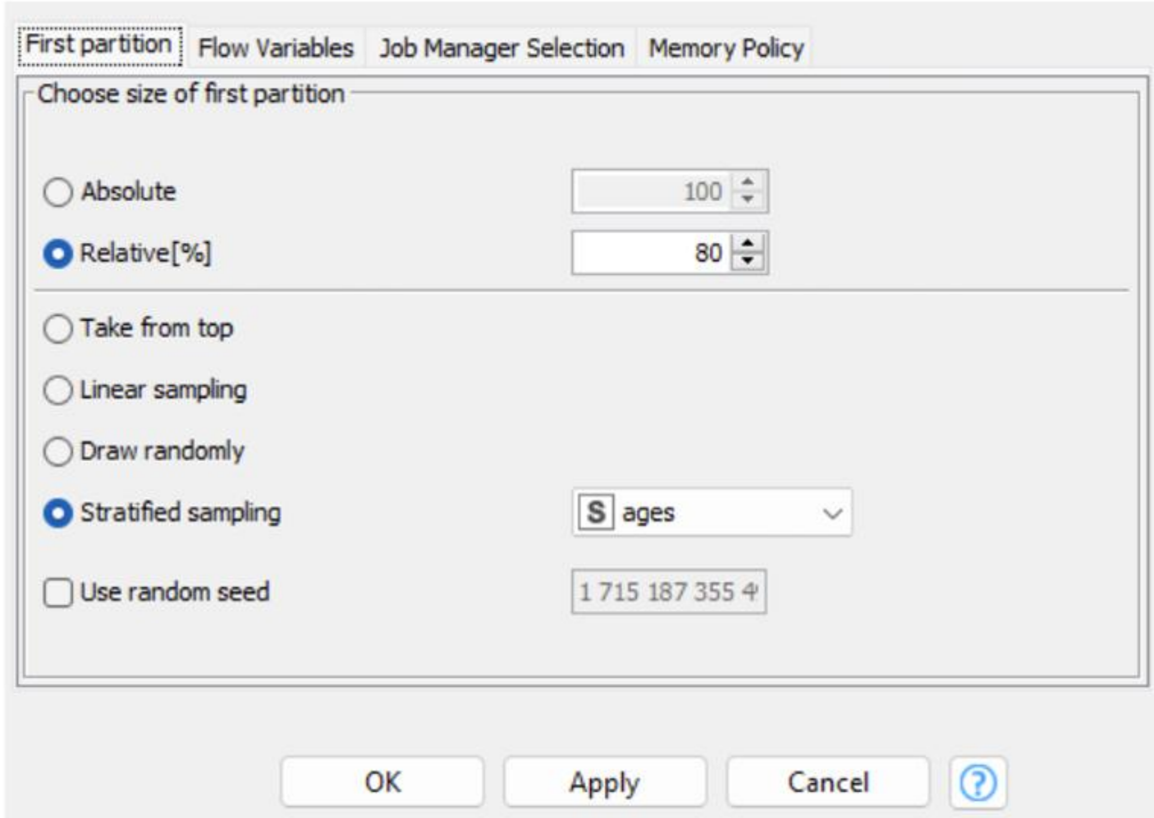
Exercise:

First Step: Partition data

Node: Partitioning



Splits a dataset into training and testing subsets



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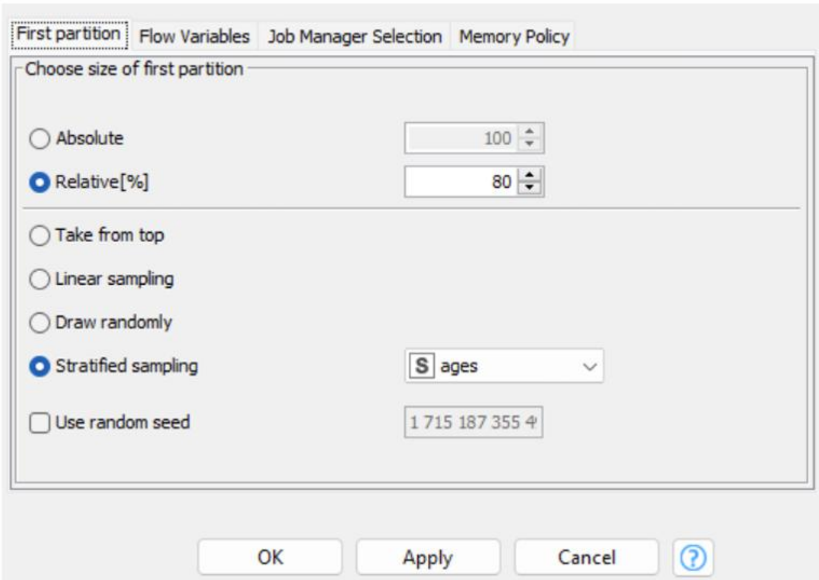
First Step: Partition data

Node: Partitioning



When setting ”relative: 80%”, it means **that 80%** of the data will be used for model **training** and the remaining **20%** will be used for **testing**.

Stratified sampling is a technique that **divides data into groups** or strata based on the values of a given variable and then randomly samples within each stratum.

When you choose ”stratified sampling: ages”, KNIME will stratify the data based on the variable ”ages” and will then sample randomly within each age group.

This is useful to ensure that the **distribution of ages is maintained** in each dataset after splitting, which can be important to avoid bias in the training and testing sets.

Make sure that the **target** is the **list_price** attribute!

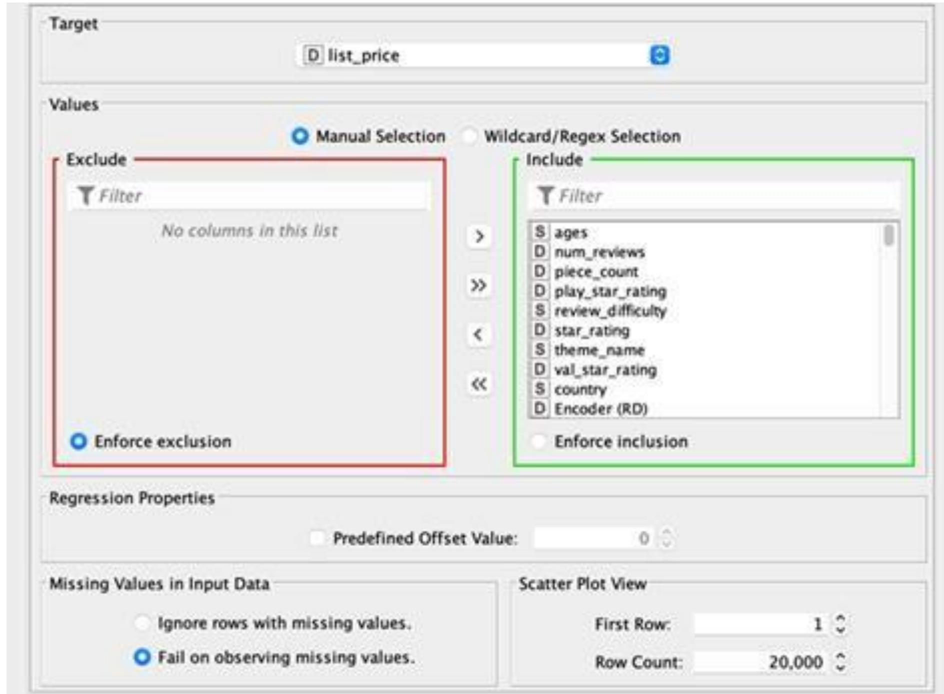
Exercise:

Second Step: Train the model

Node: Linear Regression Learner

Trains a linear regression model

Label: LEGO set price



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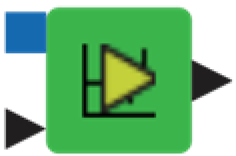
Exercise:

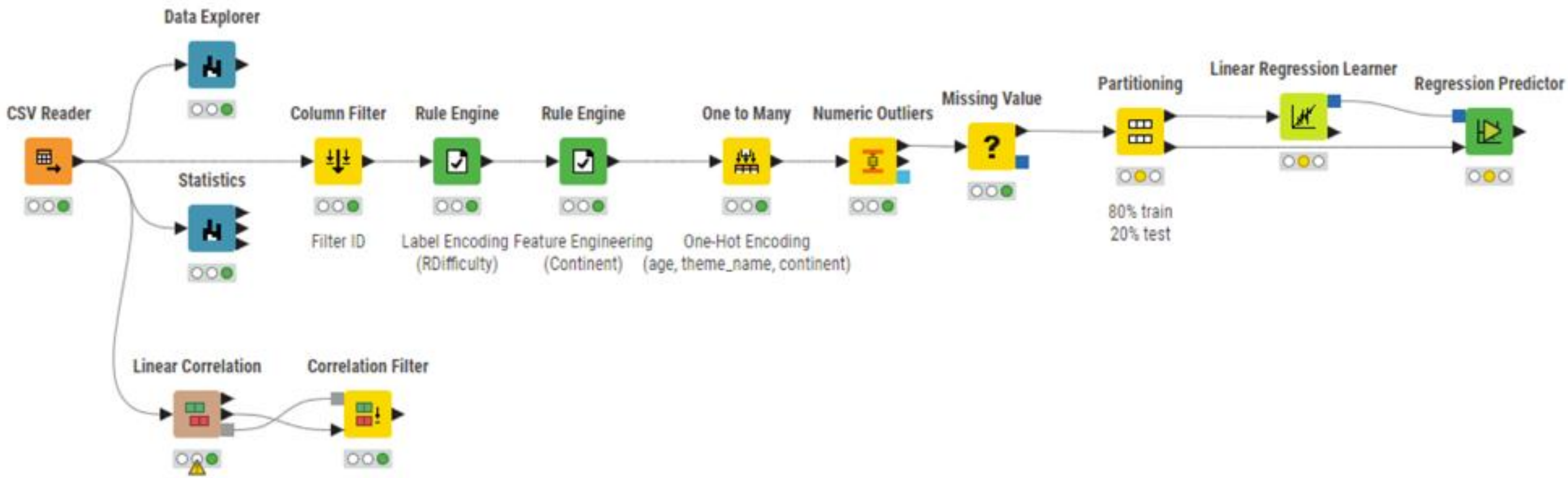
Third Step: Test the trained model

Node: Regression Predictor



- ➢ Applies a **new subset of data** (20% of the original data) to the **trained model;**
- ➢ Makes **predictions** of the LEGO set price based on the features provided in the input data.

State of the workflow



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Exercise:

Final Step: Evaluate the model

Measures:

1. **R-Square value**
2. Mean Absolute Error (MAE)
3. Mean Square Error (MSE)
4. Root Mean Square Error (RMSE)

R-Squared: proportion of variation in the result that is explained by the variables *preditors*.

$$\frac{\textit{total variance explained by model}}{\textit{total variance}}$$

Here, the higher the value, the better the model. 1 means that the model explains 100% of the variance of the labels. 0 means that the model doesn't understand how the labels vary.

Exercise:

Final Step: Evaluate the model

Measures:

1. R-Square value
2. **Mean Absolute Error** (MAE)
3. **Mean Square Error** (MSE)
4. Root Mean Square Error (RMSE)

MAE: Average of absolute value of the errors

$$\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

MSE: Mean Square Errors

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Exercise:

Final Step: Evaluate the model

Measures:

1. R-Square value
2. Mean Absolute Error (MAE)
3. Mean Square Error (MSE)
4. **Root Mean Square Error** (RMSE)

RMSE calculates the average of the square roots of the error between (real) values and predictions (hypotheses). It has a range from 0 to infinity and returns the magnitude of the errors. The scores are negatively orientated, **so lower values are better**. A score of 0 means that, on average, the forecasts are optimal, i.e. 100 per cent effective.

$$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

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Exercise:

Comparing metrics MAE, MSE and RMSE:

- **MAE** is the **easiest to understand** because it is the average error;
- **MSE** is **more popular than MAE** because MSE "punishes" larger errors, which tends to be useful in real-world problems;
- **RMSE** is **even more popular than the MSE** because the RMSE is interpretable in units“y”.

These are all loss functions, so we want to minimise them.

Exercise:

Final Step: Evaluate the model

Node: Numeric Scorer

Evaluates the performance of predictive models

➢ KNIME’s Numeric Scorer uses the predicted resource values and the actual resource values as input and **produces the metrics.**

Exercise:

Metrics Obtained:
Our model has a **R-squared value of 74.4 %**, which means that **74.4 %** of the Lego data set **lies around the regression line** created by our model.

A **MAE of 9.33** indicates that, on average, the model's predictions are **deviating about 9,33 units** from the actual values.

R^2:	0.744
Mean absolute error:	9.33
Mean squared error:	230.695
Root mean squared error:	15.189
Mean signed difference:	0.079
Mean absolute percentage error:	0.283
Adjusted R^2:	0.744

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Evaluation Metrics:

R^2 = 0.744 means we explain **74% of the price variability**.

For error, **MAE = €9.33** is the average deviation per set; **RMSE = €15.19** is similar but **penalizes large errors more** (since RMSE > MAE, we likely have a few sets with big errors).

The **mean bias is ~0**, so we're **not systematically over- or under-predicting**.

As a percentage, **MAPE ≈ 28%** is the **average relative error**.

Adjusted R^2 confirms we're **not inflating fit with useless variables**.

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Mean signed difference:	0.079
Mean absolute percentage error:	0.283
Adjusted R^2:	0.744

Useful rules

Lower is better for **MAE, MSE, RMSE, MAPE**.

Higher is better for **R^2 / Adjusted R^2**.

RMSE vs. MAE: the **bigger the gap**, the **more outlier influence**.

MAPE: avoid using it when the target can be **0 or near 0**.

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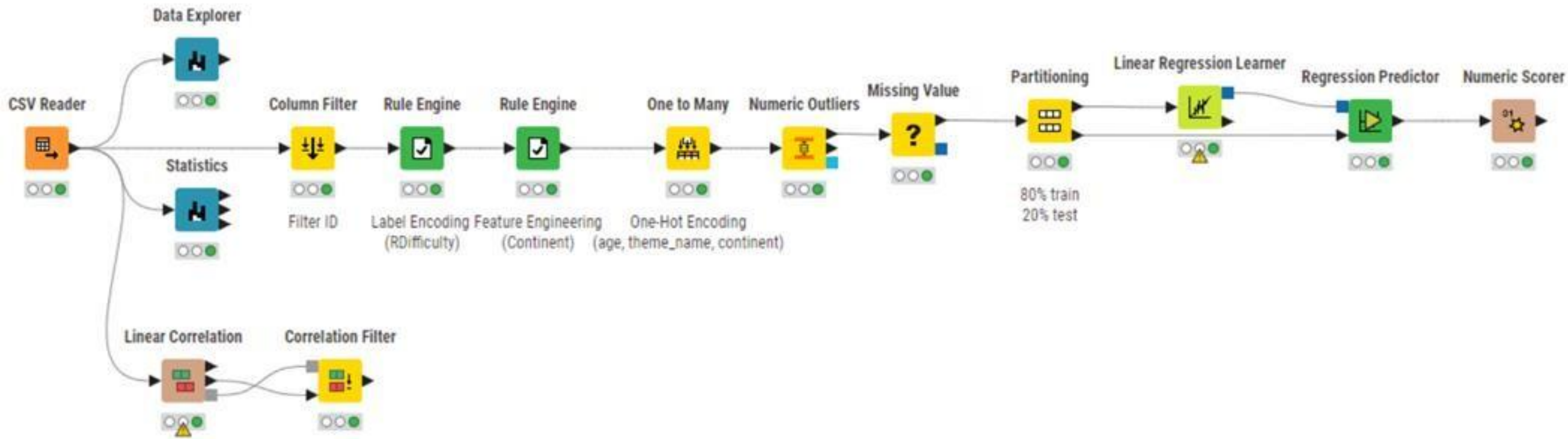
Final Workflow:



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